

AI MENTAL HEALTH

ANA – 3 (WEEK-4)

Samaritans Crisis Trends Analysis Report

Dataset Used: Office for National Statistics (ONS) suicide registrations dataset for 2024.

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1. Introduction

Mental health crises remain a major public health concern in England and Wales. Organizations such as Samaritans provide emotional support to individuals experiencing distress, loneliness, suicidal thoughts, and other mental health challenges. Understanding crisis trends helps identify vulnerable populations and regions requiring additional support services.

This project analyses publicly available suicide statistics as a proxy indicator of crisis demand. The analysis focuses on regional variation, recent trends, and changes between 2023 and 2024. Visualizations were developed to identify areas with elevated crisis risk and to support evidence-based recommendations for mental health intervention strategies.

2. Objectives

The objectives of this analysis were:

- To examine regional variation in crisis-related outcomes.
- To identify regions with the highest suicide rates.
- To compare suicide rates between 2023 and 2024.
- To evaluate recent changes in crisis risk across regions.

- To create visualizations that communicate regional trends effectively.
- To provide commentary on potential implications for crisis-support services.

3. Dataset Description

The analysis used data from the Office for National Statistics (ONS) suicide registrations dataset for 2024.

The dataset contains:

- Number of suicide deaths.
- Age-standardized suicide rates per 100,000 population.
- Regional breakdowns.
- Historical trends.
- Gender-specific statistics.
- Method-of-death information.

The primary source for this analysis was Table 10, which provides regional suicide rates for England and Wales.

4. Data Preprocessing

Several preprocessing steps were undertaken before analysis:

1. Imported the Excel dataset into Python.
2. Loaded regional statistics from Table 10.
3. Filtered records to include only "Persons".
4. Removed national totals to focus on regional comparisons.
5. Selected suicide rate variables for 2023 and 2024.

6. Calculated year-on-year rate changes.
7. Prepared cleaned datasets for visualization and reporting.

5. Data Analysis

5.1 Regional Crisis Risk Analysis

Regional suicide rates for 2024 were analyzed to identify geographical differences in crisis-related outcomes.

Regions were ranked according to age-standardized suicide rates per 100,000 population.

Highest-Risk Regions

- Wales
- North East
- North West
- Yorkshire and Humber
- East Midlands

These regions demonstrated substantially higher suicide rates than the national average.

Lowest-Risk Regions

- London
- East of England
- West Midlands

These areas recorded comparatively lower rates.

	Region	Rate_2023	Rate_2024
0	England	11.9	11.8
1	North East	15.9	16.0
2	North West	16.0	14.8
3	Yorkshire and The Humber	12.8	14.4
4	East Midlands	12.6	12.9

5.2 Year-on-Year Change Analysis

A comparison between 2023 and 2024 was conducted to identify regions where crisis risk increased or decreased.

The change was calculated as:

Change = Rate(2024) – Rate(2023)

Positive values indicate increasing crisis risk, while negative values indicate improvement.

index	Region	Rate_2023	Rate_2024	Change
10	Wales	14.4	16.2	1.7999999999999999
1	North East	15.9	16.0	0.09999999999999996
2	North West	16.0	14.8	-1.1999999999999993
3	Yorkshire and The Humber	12.8	14.4	1.5999999999999996
4	East Midlands	12.6	12.9	0.30000000000000007

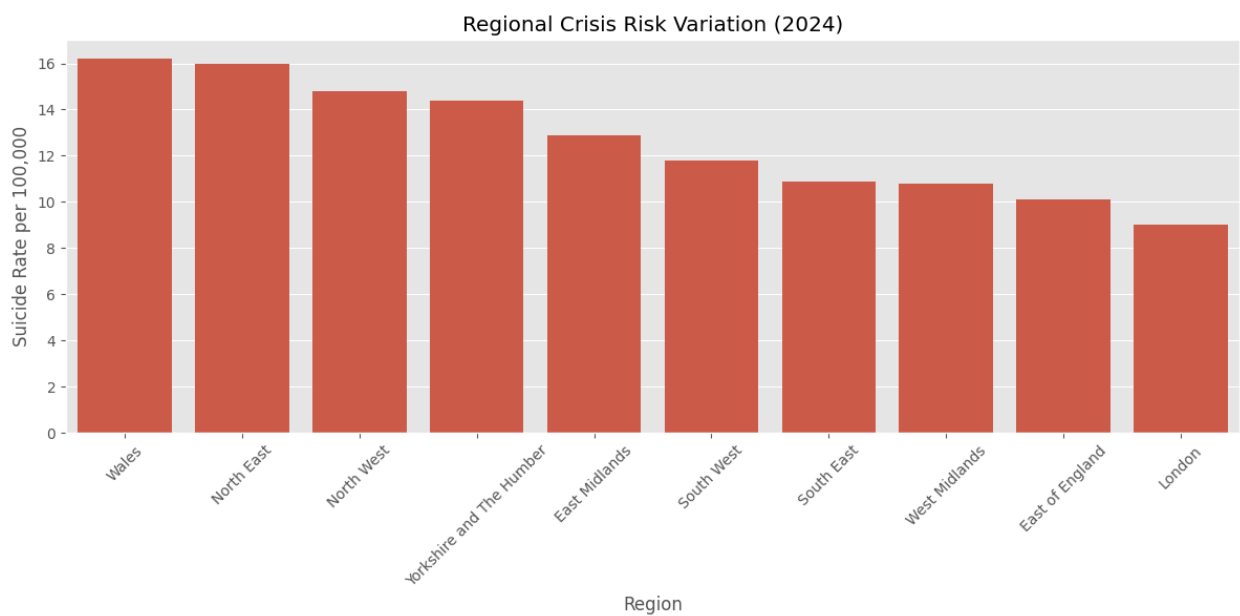
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6. Data Visualization

6.1: Regional Crisis Risk Variation (2024)

The first visualization presents regional suicide rates for 2024.

The chart highlights significant geographical variation in crisis-related outcomes. Wales and the North East exhibited the highest rates, while London reported the lowest.

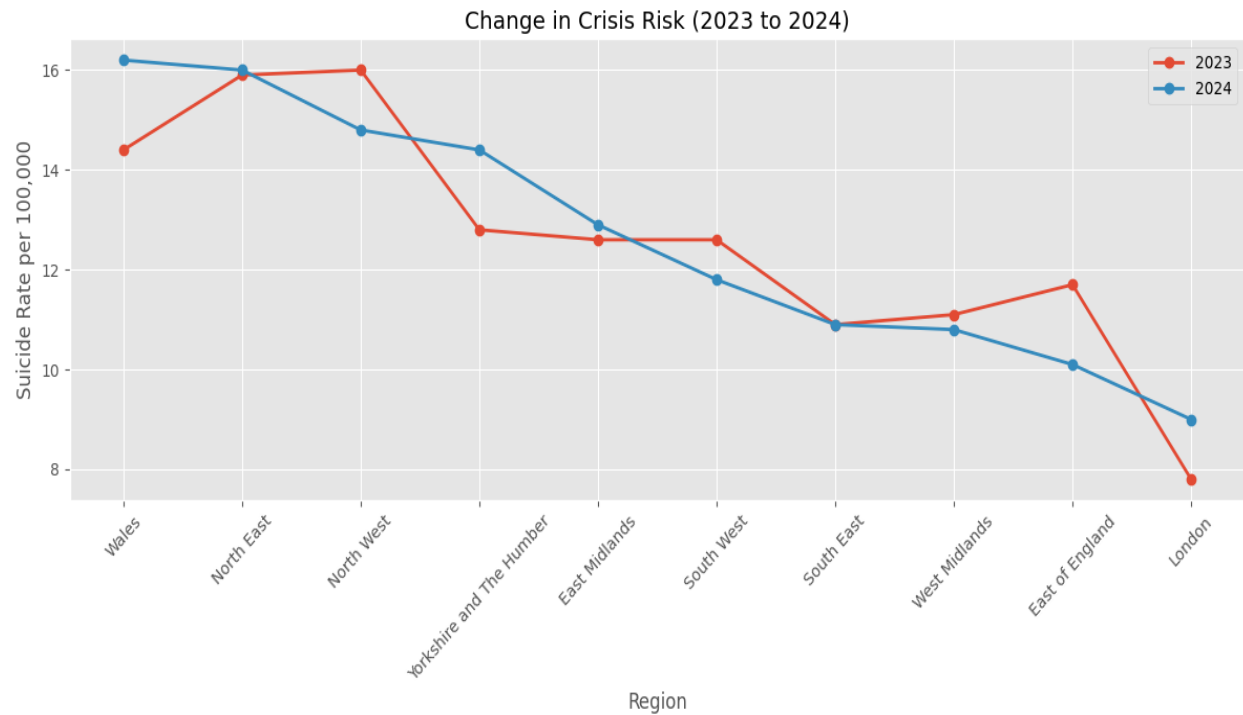


Interpretation

The variation suggests unequal distribution of crisis burden across regions. Areas with persistently elevated rates may require enhanced crisis intervention services and mental health support infrastructure.

6.2: Change in Crisis Risk (2023–2024)

A line chart was developed to compare regional suicide rates between 2023 and 2024.



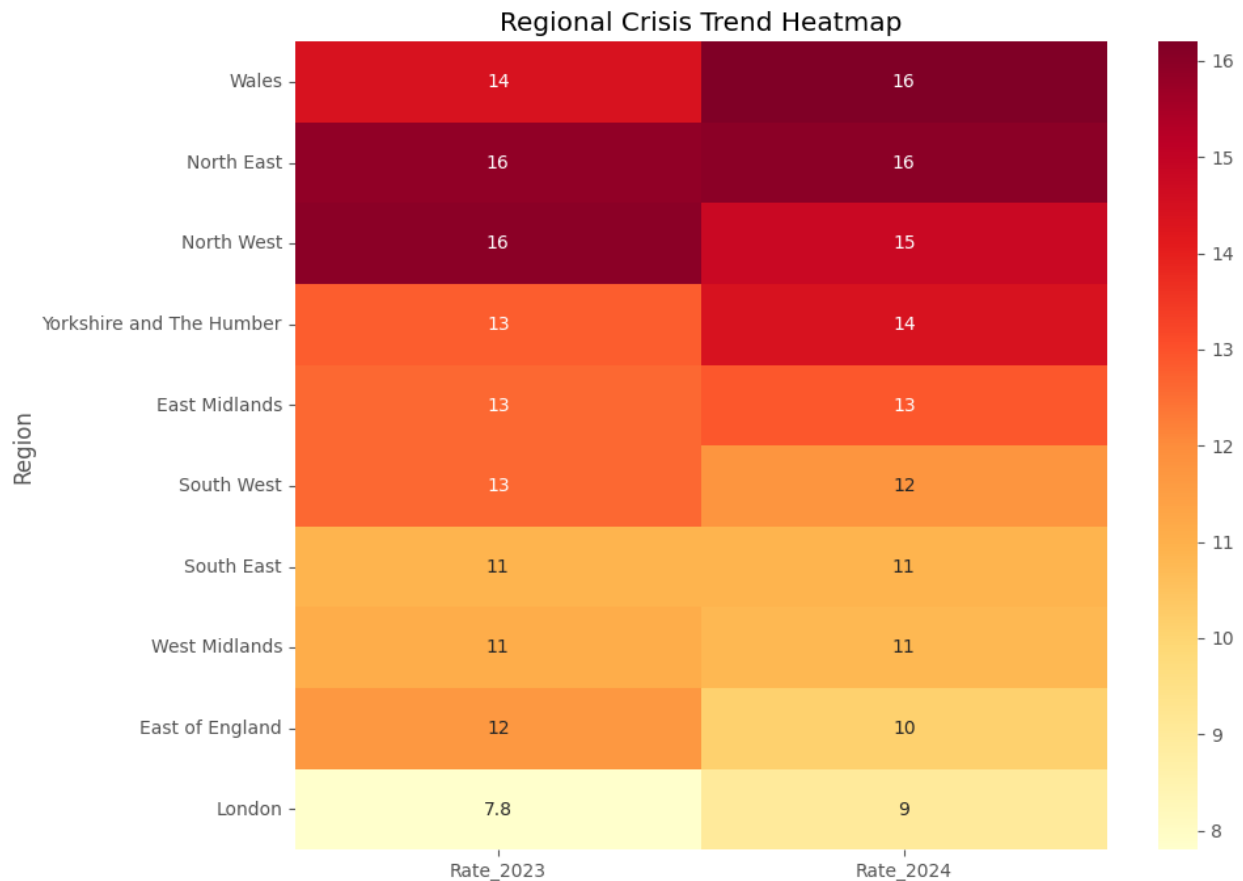
Interpretation

The visualization allows direct comparison of regional performance across the two years and highlights areas where rates have increased.

Regions showing upward movement may indicate worsening mental health pressures, socioeconomic challenges, or gaps in support provision.

6.3: Regional Crisis Trend Heatmap

A heatmap was produced to visualize relative crisis intensity across regions.



Interpretation

Darker areas represent higher suicide rates and enable rapid identification of high-risk regions.

The heatmap provides an intuitive overview of geographical disparities and supports prioritization of intervention resources.

7. Key Findings

The analysis produced several important findings:

7.1: Significant Regional Variation

Suicide rates varied substantially across England and Wales, indicating differences in crisis burden and mental health outcomes.

7.2: Wales Recorded the Highest Crisis Risk

Wales reported the highest age-standardized suicide rate among all regions examined.

7.3: Northern Regions Remain High Risk

The North East and North West consistently exhibited elevated rates compared with many southern regions.

7.4: London Recorded the Lowest Rate

London maintained the lowest suicide rate, suggesting comparatively better outcomes or differing demographic and socioeconomic characteristics.

7.5: Evidence of Continuing Crisis Pressures

Several regions showed increases between 2023 and 2024, indicating that crisis pressures remain a significant public health challenge.

7.6: Persistent Geographical Inequalities

The observed regional differences suggest that crisis experiences and support needs are not evenly distributed throughout the country.

8. Conclusion

This study examined crisis-related trends using regional suicide statistics for England and Wales. The findings revealed substantial regional variation, with Wales and several northern regions experiencing the highest levels of crisis

risk. Comparison of 2023 and 2024 data identified areas where rates have increased, suggesting ongoing mental health pressures.

The visualizations and statistical analysis provide valuable evidence for policymakers, healthcare organizations, and crisis-support services such as Samaritans. Continued monitoring of regional trends is essential for improving crisis intervention strategies and reducing mental health inequalities.